

Name: _____

Double Award Science – Physics Unit 2 Higher Tier

Once you have completed a paper and have marked it, fill out the table below, using the grade boundaries provided to add your grade. Then list which questions you did well and which questions you struggled with. For the questions you struggled with, review why and at the reasons. The most common reasons are

- Not understanding basic concepts (e.g. simple definitions)
- Calculation errors (forgetting to square numbers, etc.)
- Not reading questions carefully
- Not answering the full question (describe vs. explain)
- Laboratory/practical setup mistakes

For these questions it is also useful add notes to the question to help you next time and take a photo of these questions. Use these then to guide further revision.

Year	Score /60	Grade	Good Questions	Bad Questions	Reasons
2017					
2018					
2019					
2022					
2023					

Grade boundaries - Raw Score

Year	E	D	C	B	A	A*	Max
2017	2	6	13	20	28	36	44
2018	3	5	10	17	35	33	41
2019	6	9	15	21	28	35	42
2022	3	5	11	18	26	34	42
2023	6	10	16	23	34	38	45

Surname	Centre Number	Candidate Number
Other Names		0



GCSE – **NEW**

3430UF0-1



S18-3430UF0-1

SCIENCE (Double Award)

Unit 6 – PHYSICS 2
HIGHER TIER

WEDNESDAY, 23 MAY 2018 – AFTERNOON

1 hour 15 minutes

For Examiner's use only		
Question	Maximum Mark	Mark Awarded
1.	15	
2.	17	
3.	9	
4.	9	
5.	10	
Total	60	

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01

ADDITIONAL MATERIALS

In addition to this examination paper, you may require a calculator and a ruler.

INSTRUCTIONS TO CANDIDATES

Use black ink or black ball-point pen. Do not use gel pen. Do not use correction fluid.
Write your name, centre number and candidate number in the spaces at the top of this page.
Answer **all** questions.
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INFORMATION FOR CANDIDATES

The number of marks is given in brackets at the end of each question or part-question.
The assessment of the quality of extended response (QER) will take place in question **3(b)**.



MAY183430UF0101

Equations

speed = $\frac{\text{distance}}{\text{time}}$	
acceleration [or deceleration] = $\frac{\text{change in velocity}}{\text{time}}$	$a = \frac{\Delta v}{t}$
acceleration = gradient of a velocity-time graph	
distance travelled = area under a velocity-time graph	
resultant force = mass $\times$ acceleration	$F = ma$
weight = mass $\times$ gravitational field strength	$W = mg$
work = force $\times$ distance	$W = Fd$
kinetic energy = $\frac{\text{mass} \times \text{velocity}^2}{2}$	$\text{KE} = \frac{1}{2}mv^2$
change in potential energy = mass $\times$ gravitational field strength $\times$ change in height	$\text{PE} = mgh$
force = spring constant $\times$ extension	$F = kx$
work done in stretching = area under a force-extension graph	$W = \frac{1}{2}Fx$

SI multipliers

Prefix	Multiplier
p	1×10^{-12}
n	1×10^{-9}
μ	1×10^{-6}
m	1×10^{-3}

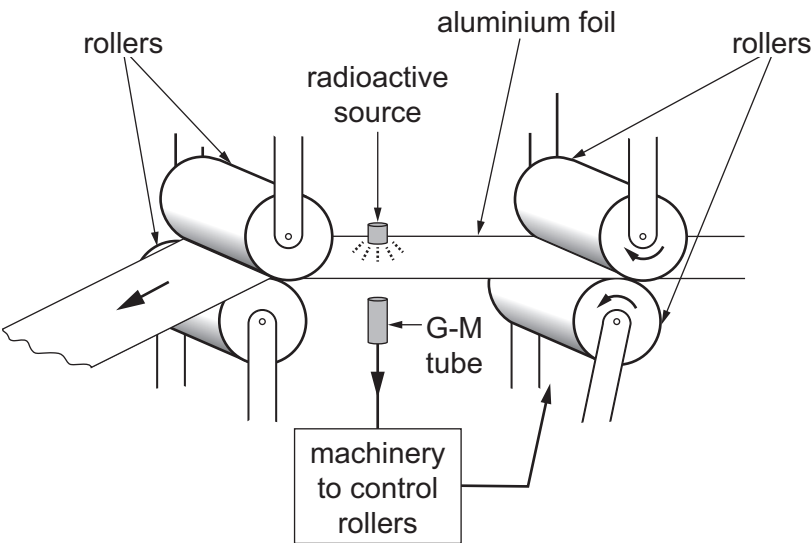
Prefix	Multiplier
k	1×10^3
M	1×10^6
G	1×10^9
T	1×10^{12}



Answer all questions.

1. A group of students study the uses of radioactivity. They find that radioactive isotopes are widely used in a variety of applications. For example, alpha emitters are used in smoke detectors. Medical diagnosis and cancer treatments use a range of radioisotopes emitting alpha, beta and gamma radiation.

One use they study in detail is monitoring the thickness of aluminium foil when it is manufactured. Radiation passes through the aluminium foil and is detected by a G-M tube. Changes to the thickness cause a difference in the count rate detected and adjustments can then be made to the pressure applied by the rollers.



Different radioisotopes have different half-lives and decay in different ways. The properties of some radioisotopes are given in the table below.

Isotope	Symbol	Half-life	Decay mode
strontium-90	$^{90}_{38}\text{Sr}$	29 years	beta
americium-241	$^{241}_{95}\text{Am}$	432 years	alpha
caesium-137	$^{137}_{55}\text{Cs}$	30 years	gamma
phosphorous-32	$^{32}_{15}\text{P}$	14 days	beta
actinium-225	$^{225}_{89}\text{Ac}$	10 days	alpha



(a) Tick (✓) the boxes alongside the **three** correct statements below. [3]

- Alpha radiation consists of helium nuclei ☐
- Alpha radiation is more ionising than gamma ☐
- Strontium-90 has 90 nucleons ☐
- Gamma radiation consists of low energy waves ☐
- Gamma radiation only travels a short range in air ☐
- Beta radiation consists of slow moving electrons ☐

(b) One of the students suggests that strontium-90 is the most suitable isotope from the table for monitoring the thickness of aluminium foil. Explain whether or not you agree with this suggestion. [3]

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(c) (i) Explain what is meant by the statement: ‘The half-life of strontium-90 is 29 years.’ [2]

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(ii) Calculate the time taken for the activity of strontium-90 to fall to $\frac{1}{8}$ th of its initial value. [3]

Time = years

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(d) The students' teacher demonstrates experiments with radioactive sources. First she measures the radiation in the laboratory, recording 150 counts in 5 minutes. This allows her to work out the count rate of the background radiation.

(i) Calculate the background radiation count rate in counts per **second** (cps). [2]

Background count rate = cps

(ii) Suggest **two** ways in which the teacher could improve the accuracy of her result. [2]

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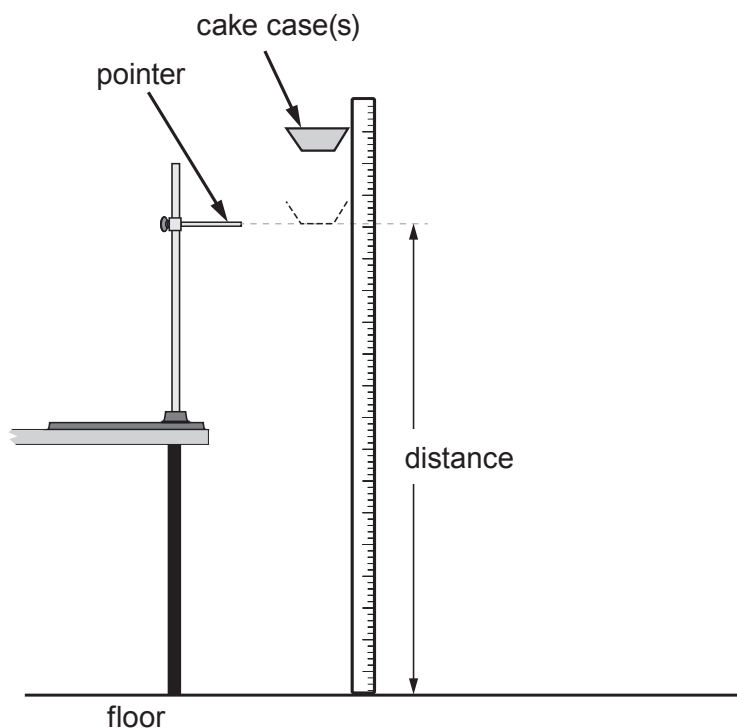
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2. A group of students investigate if the terminal speed of falling paper cake cases depends on their mass. They follow the method given below.



1. Set up a pointer in the clamp stand and set it 1.50 m above the ground.
2. Take a single cake case and record its mass using a balance.
3. Drop the cake case from 20 cm above the pointer.
4. Use a stopwatch to record the time it takes to fall from the level of the pointer to the floor.
5. Repeat steps 3 and 4 another four times.
6. Repeat steps 3-5 with extra cake cases in a stack.

- (a) (i) Give **one** reason why the students let the cake case fall for 20 cm before starting the stopwatch. [1]

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- (ii) Give **two** reasons why the students take more than one repeat reading. [2]

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(b) The following data are collected. The students assume that each of the cake cases has a mass of 0.5g.

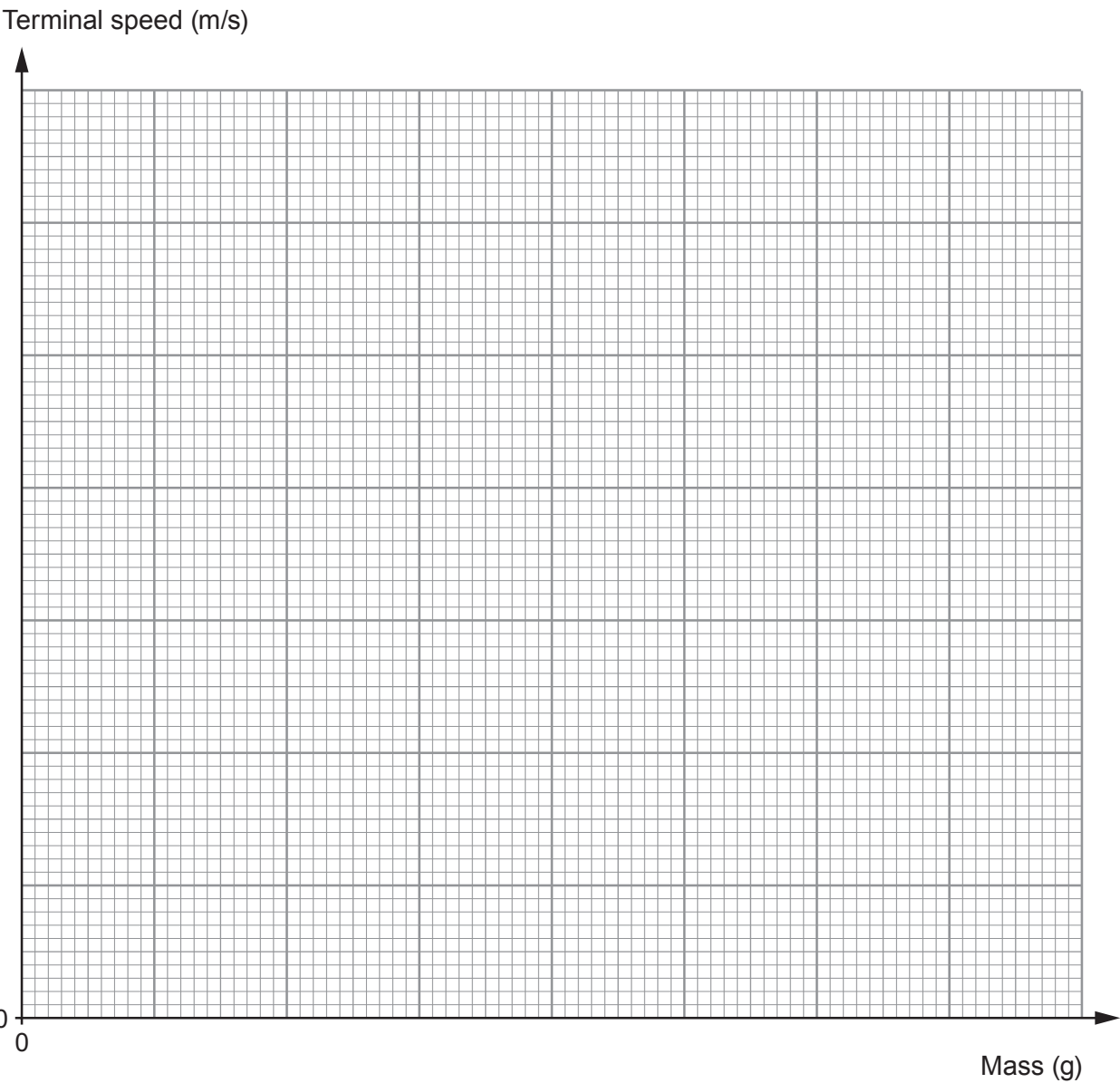
Number of cake cases	Mass of cake cases (g)	Mean time taken for cake cases to fall 1.50 m (s)	Terminal speed (m/s)
0	0		0
1	0.5	0.94	1.60
2	1.0	0.67	2.24
3	1.5	0.59	2.54
5	2.5		2.88
6	3.0	0.51	2.94

(i) **Complete the table.** Use an equation from page 2 to calculate the missing value of the mean time. *Space for workings.* [2]



(ii) Plot the data on the grid below and draw a suitable line.

[4]



(iii) One of the students suggests that the terminal speed will always increase by a factor of 1.4 if the mass is doubled. The student finds that when the mass doubles from 0.5g to 1.0g this suggestion is true. Explain if this is true for the other masses. [3]

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- (c) Apart from taking more repeat readings, suggest one way in which the method could be improved to collect better quality data **and** explain how the improvement would give better data. [2]

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- (d) Explaining your reasoning and using an equation from page 2, calculate the size of the air resistance force acting on a stack of 5 cake cases when travelling at terminal speed. (Gravitational field strength, $g = 10 \text{ N/kg}$). *Space for workings.* [3]

Air resistance = N

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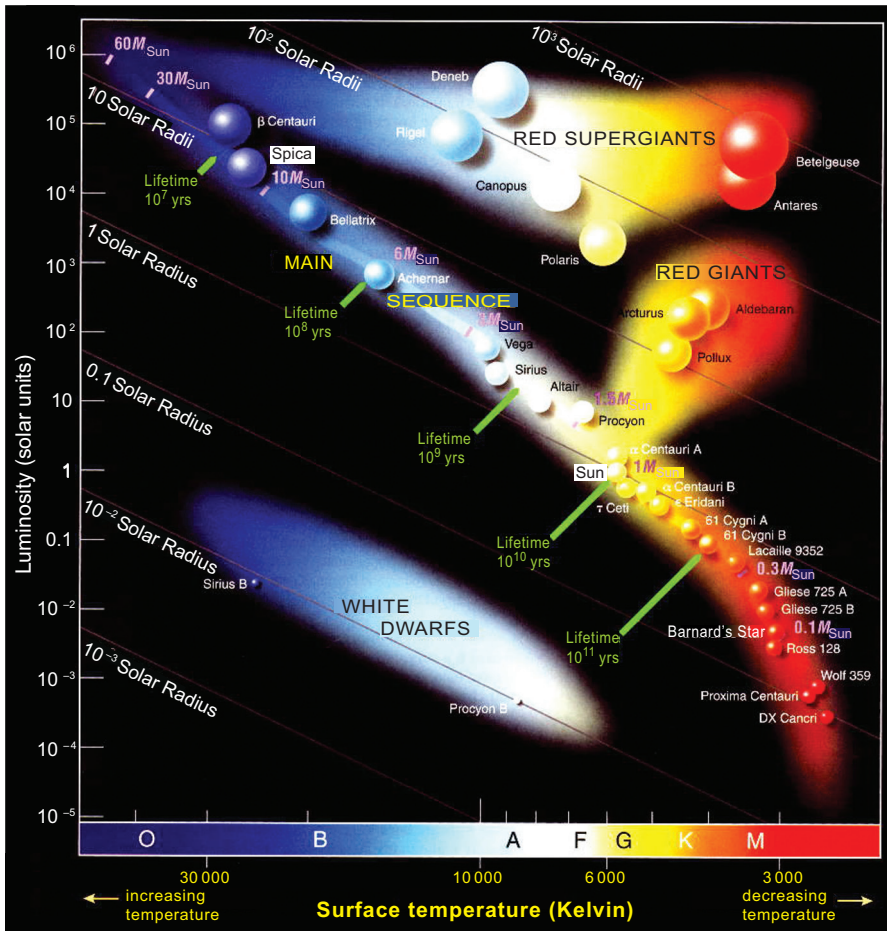
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3. One of the most useful diagrams in astrophysics is the Hertzsprung-Russell (HR) diagram. It was developed separately by two astronomers, the Danish astronomer Ejnar Hertzsprung and the American astronomer Henry Norris Russell. Their combined diagram shows that the relationship between the temperature and luminosity of a star is not random and stars fall into distinct groups. These are seen in the Hertzsprung-Russell diagram below which shows the properties of stars.



- (a) Spica is a very large blue star on the main sequence, approximately 10 times the mass of the Sun. Spica is 261 light-years from Earth.
- (i) Explain, in terms of forces, why Spica is currently stable. [2]
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- (ii) Spica is 261 light-years from Earth. State what this means. [1]
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(b) The Sun is also on the main sequence. It is a yellow dwarf star. Explain in terms of forces, the reactions within the Sun and the properties shown on the HR diagram how the Sun will change during the remainder of its life cycle. [6 QER]

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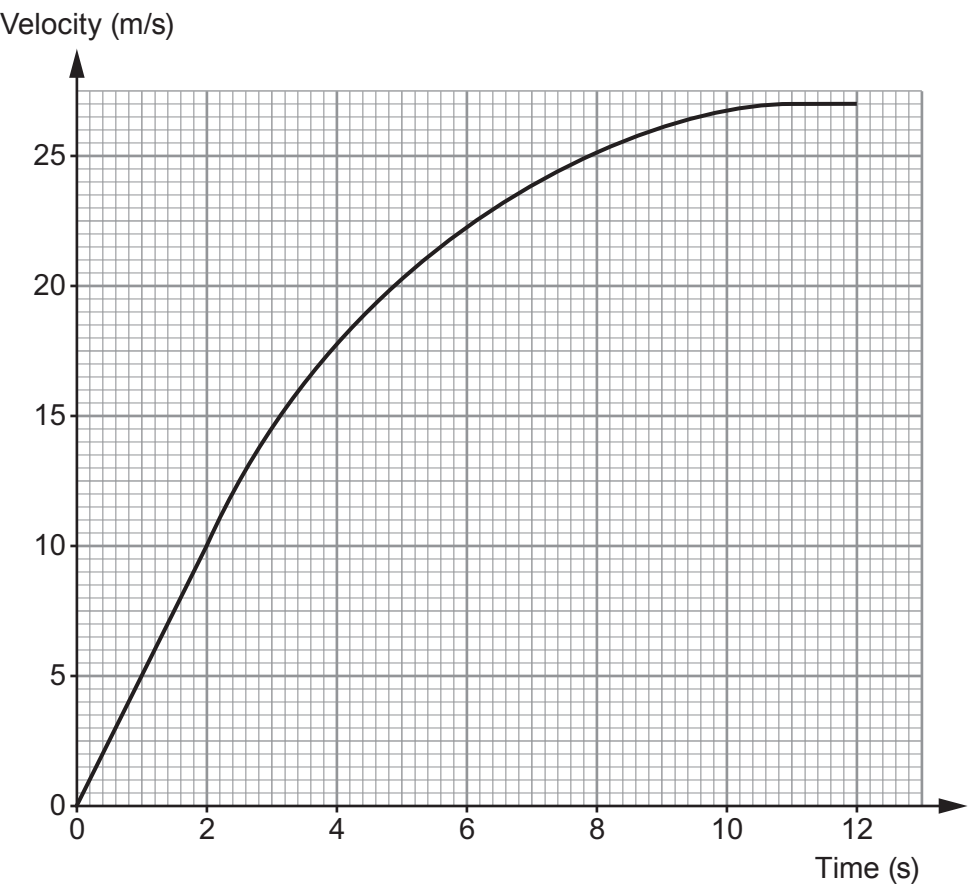
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4. Manufacturers test new cars on a level track. In order to find out how long it takes them to accelerate to 27 m/s (60 mph), the cars are driven in a straight line at maximum power and the speed recorded. Data for one car is shown on the graph below.



(a) Describe how the **acceleration** changes during the 12 s shown. [2]

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(b) By drawing a suitable tangent and using:
acceleration = gradient of a velocity-time graph
calculate the acceleration of the car at 5 s. Give a unit with your answer. [3]

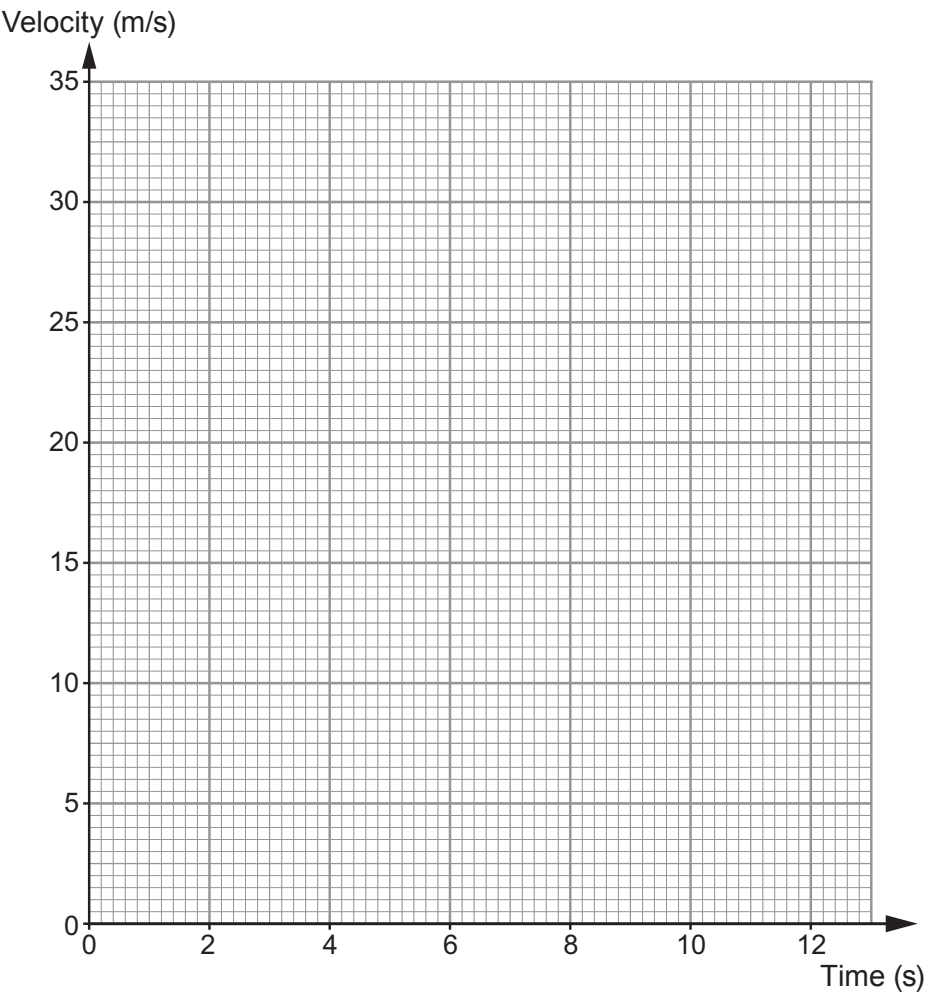
Acceleration = Unit =

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(c) Use an equation from page 2 to estimate the distance travelled by the car in the first 3 s. [2]

Distance = m

(d) Another car of the same power and mass but with a more streamlined shape is tested. **Sketch on the grid below** the velocity-time graph for this car. [2]



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5. Almost all cars rely on fossil fuels as their source of energy. However with supplies of fossil fuels decreasing and concerns over climate change increasing, it is important to make cars as energy efficient as possible. One type of car which is very efficient is a hybrid electric vehicle, which has both a conventional engine and an electric motor which runs from batteries. These hybrid vehicles use regenerative braking where some of the kinetic energy transferred when the brakes are applied is used to charge the batteries rather than all being lost as heat. This has a maximum energy transfer of 60%.

Data about a hybrid electric / petrol car is given below.

Time taken to accelerate from 0-100 km/h (0-28 m/s)	11.8 s
Mass of car	1 160 kg
Mean CO ₂ emissions	8.9×10^{-2} kg/km
Mean fuel economy	31.6 km/litre

- (a) The car uses 10 litres of fuel per week. Calculate the mean mass of CO₂ emitted by the car in a year (52 weeks). [3]

Mean mass of CO₂ = kg

- (b) Energy can be lost from cars in a variety of ways such as inertial losses, rolling resistance losses and idling losses. State the design features that manufacturers can use to reduce **each** of these energy losses. [3]

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- (c) During a journey the car slows down from 30 m/s to 10 m/s.
Use an equation from page 2 to calculate how much electrical energy is transferred to the battery if the regenerative braking process is at its maximum energy transfer. [4]

Energy transferred = J

10

END OF PAPER



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GCSE

3430UF0-1



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Prefix	Multiplier
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Answer all questions.

1. (a) Pupils in a class were given 200 coins to use in an experiment to simulate radioactive decay. They were asked to shake the coins in a bag, throw them out on the table then remove those showing “heads”. The number removed were counted and recorded in a table. The remainder of the coins were put back in the bag and the process was repeated again and again. Their results are shown in the table below.

Throw number	Total number of coins removed	Number of coins remaining
0	0	200
1	104	96
2	149	51
3		26
4		20
5		6
6		4

- (i) **Complete the table.** [1]
- (ii) After two throws, the number of coins remaining was 51. How many coins would you have **expected** to remain? [1]
-
- (iii) After how many throws would the number of remaining coins fall to about **one eighth** of the original number? [1]
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- (b) Carbon-14 is a radioactive form of carbon that is present in all living material. Each nucleus of carbon-14 undergoes radioactive decay by emitting a beta particle to form nitrogen-14 according to the following decay equation, which is incomplete.

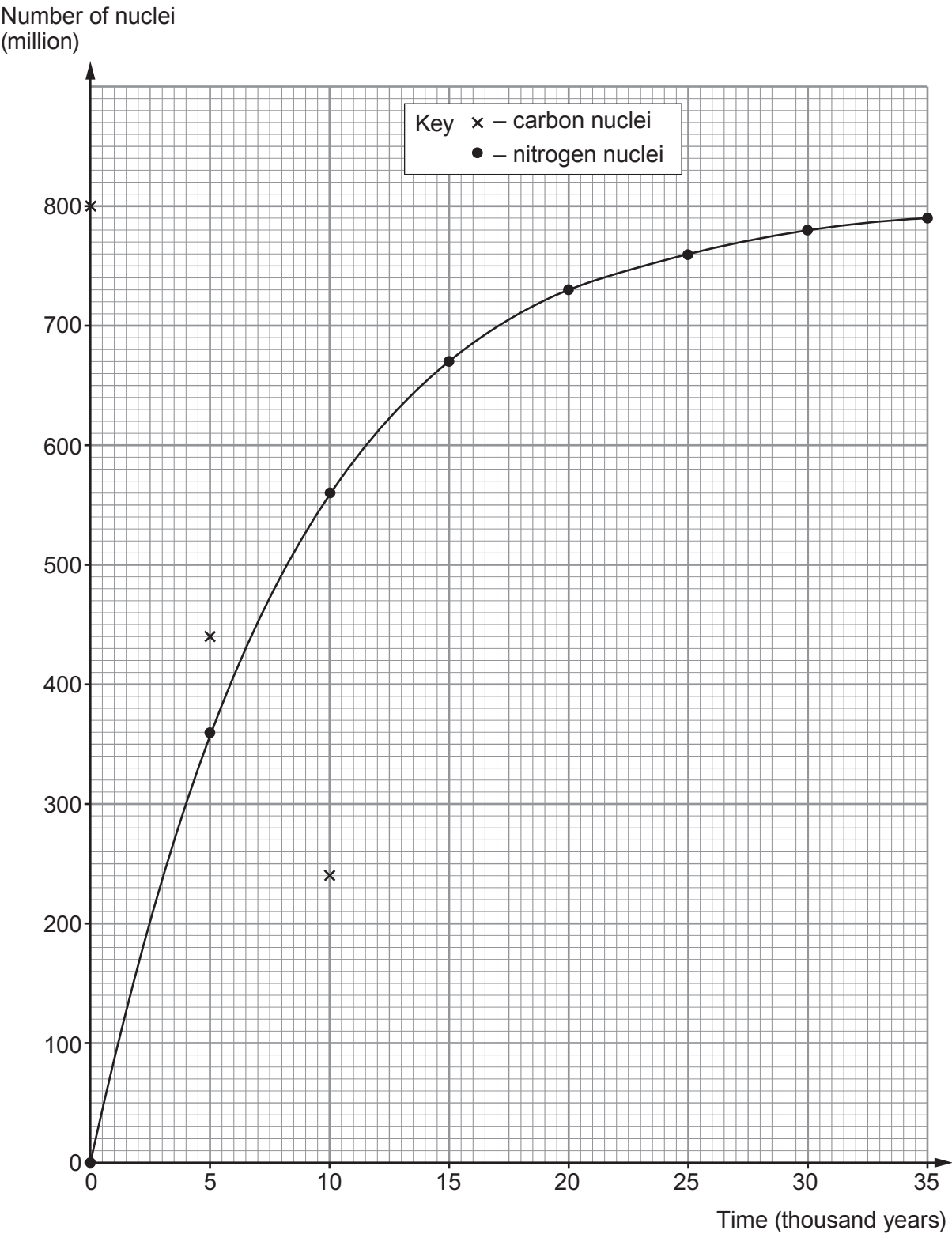


Complete the nuclear equation above.

[2]



- (c) A sample of 800 million carbon nuclei decays to create nuclei of nitrogen. The decrease in the sample creates an **increase** in the number of nuclei of nitrogen according to the following graph.



(i) Complete the following table. [2]

Time (thousand years)	Total number of nuclei (million)	Number of nitrogen nuclei (million)	Number of carbon nuclei (million)
0	800	0	800
5		360	440
10		560	240
15		670	
20		730	
25		760	
30		780	
35		790	

(ii) On the grid opposite, plot points showing the **decay** of the carbon-14 nuclei. The first three crosses showing the numbers of carbon nuclei have been plotted for you.
Draw a suitable line. [3]

(d) (i) State the meaning of “the half-life” of a radioactive substance. [2]

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(ii) Use the graph to determine the half-life of carbon-14. [1]

Half-life = thousand years

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- (e) Carbon dating is used to find the age of some ancient objects because carbon-14 is present in all once-living material. The process has been used to identify the age of the Turin shroud. This is a sheet of cloth that was claimed to be about 2000 years old. Three independent radiocarbon dating tests, carried out recently, attempted to identify the age of the cloth.



Out of **80 million** carbon-14 nuclei which were present in each sample of the original cloth, around 6 million have decayed into nitrogen. Use this information to explain whether the claim about its age is correct. [2]

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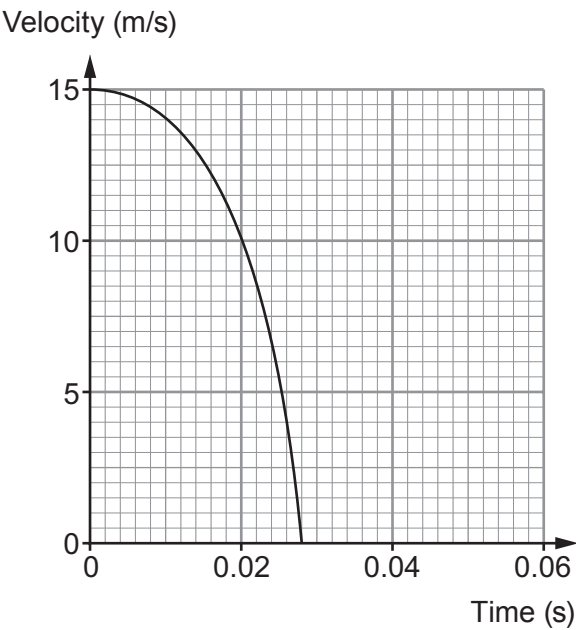
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2. The safety of car drivers and passengers in the event of a collision has been the subject of an enormous amount of research in the last 50 years. As a result, the design of cars has developed to improve the safety of the occupants in the event of a head-on collision.

A crash dummy is positioned in the driving seat of a car, which is directed, under test conditions, into a solid concrete wall. The graph below shows the velocity of the dummy from the moment the car makes contact with the wall.



- (a) State Newton’s third law of motion and explain how it applies to the car and the wall in this collision. [3]

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- (b) (i) Use the graph opposite and equations from page 2 to calculate the mean resultant force on the dummy of mass 85 kg during the collision.
Give your answer to two significant figures. [3]

Mean resultant force = N

- (ii) Use the graph opposite to explain whether the resultant force is constant during the collision. [2]

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- (c) Explain, **in terms of the work done**, how a crumple zone at the front of the car would improve the safety of the driver in a head-on collision. [2]

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3. (a) Explain, in terms of forces and nuclear fusion, the life cycle of a high-mass star, from its position on the main sequence to its “death”. [6 QER]

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(b) The number of planets in our Solar System totals eight.
Briefly outline the difference between the structure of the inner and outer four planets and explain how they originated. [3]

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4. Nuclear waste materials give out alpha, beta and gamma radiations and are classified according to their level of activity. They fall into three groups as follows:

Group	Information
Low level waste	Contain relatively low levels of activity and consist of alpha emitters and low activity beta and gamma emitters.
Intermediate level waste	Have higher levels of activity than low level waste but are not as dangerous as high level waste.
High level waste	Produce a significant temperature rise as a result of their high activity. This has to be considered when planning waste storage or disposal methods.

(a) Outline the dangers to people’s health from each of alpha, beta and gamma radiations from sources that are external to the body. [3]

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(b) Low level waste can, after some basic treatment, be dumped in approved local authority sites. However, intermediate and high level wastes pose a much larger threat to life. They must be stored or disposed of in a way that keeps them well away from life forms for a very long time, typically tens of thousands of years or more.

The table below gives some data about four countries that generate significant amounts of high level waste from nuclear power stations.

Country	U.K.	Sweden	U.S.A.	France
Land area (km ²)	241 930	407 310	9 147 420	547 557
Population	65 637 240	9 903 120	323 127 510	66 689 000
Mean population per km ²	271.7	24.3	35.3	121.8
Number of nuclear reactors	15	8	99	58
Volume of low level waste in storage (m ³)	30 100	No information available	52 800	87 000
Volume of intermediate level waste in storage (m ³)	99 000		416 060	40 000
Volume of high level waste in storage (m ³)	1960			2 700

(i) A person has suggested that for the U.K. and France, the volume of low level waste increases in proportion to the number of nuclear reactors in the country.
Explain whether the suggestion is valid. [2]
Space for calculations.

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- (ii) Sweden, U.S.A. and France are well advanced in their planning for the long term storage of intermediate and high level wastes in deep underground storage facilities.

In 2010, the scientific director of the French authority for the long term storage of nuclear waste visited the U.S.A. storage site at Yucca Valley. Following his visit, he made the statement “There are far better sites in the U.S.A. than at Yucca mountain. When I went to the top of Yucca mountain and saw the volcanoes below, that worried me”.

The development of the Yucca site has been stopped since around that time despite the spending of \$10 billion on two decades of work. The Americans are still investigating an alternative site to Yucca Valley.

- I. Explain the reason for concern over the choice of the Yucca Valley for the long term storage of nuclear waste. [2]

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- II. Use information in the table opposite to suggest a reason why a country such as the U.K. finds it difficult to agree on a site for a deep storage facility. [2]

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- (iii) The U.K. opened the world’s first nuclear power station in 1956 in Calder Hall in Windscale amidst claims that, in the future, electricity would be free. The present view is that, with everything taken into account, electricity production from nuclear fuel is unacceptably expensive. Explain why there has been a change of view. [2]

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- (iv) Discuss an advantage and a disadvantage of **one** option, other than deep underground storage, for the long term disposal of dangerous nuclear waste. [2]

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5. Springs are used in a wide variety of applications, ranging from those used in the suspension systems of heavy vehicles to those used in retracting ball-point pens.

The table below shows the properties of a number of springs that are available from a particular supplier.

Spring	Coil diameter (mm)	Wire diameter (mm)	Maximum load (N)	Unstretched length (mm)	Spring constant (N/mm)	Maximum length (mm)
A	16.00	0.165	1.252	6.350	0.17	13.716
B	16.00	0.165	1.031	6.350	0.14	13.716
C	16.00	0.165		6.350	0.12	17.050
D	16.00	0.165	1.257	9.524	0.09	23.495
E	16.00	0.230	1.768	7.950	0.24	14.016
F	16.00	0.230	1.524	7.950	0.12	20.650

- (a) (i) Write down the spring that can withstand a maximum load of at least 1.2 N and has a maximum extension close to 14 mm. [1]

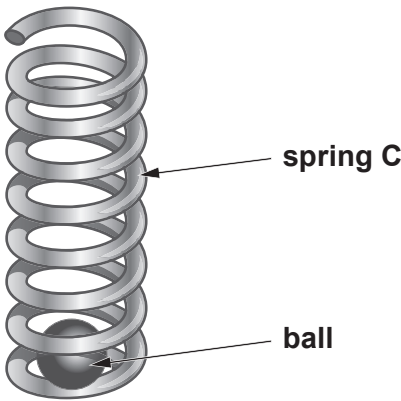
Spring =

- (ii) Express the spring constant for **spring A** in standard units of N/m. [1]

Spring constant, k = N/m



- (b) A group of students want to use a spring to project a small ball vertically upwards. They choose to use **spring C**. The bottom of their spring would hold the ball within it and the top of the spring would be fixed in position. The bottom of the spring would be pulled down and released to project the ball vertically upwards.



- (i) Use equations from page 2 to calculate the energy stored in the spring when it is stretched to its maximum length. [3]

Stored energy = J

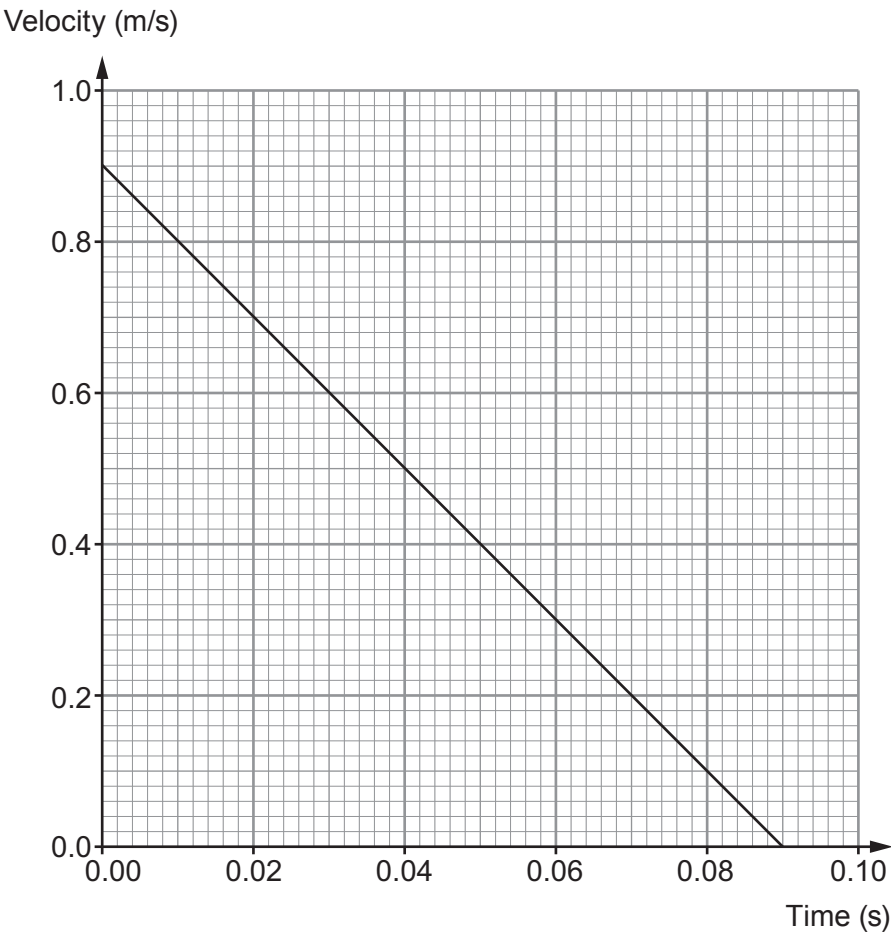
- (ii) Use an equation from page 2 to calculate the maximum theoretical height that the ball would reach given that its mass is 15 grams. [3]
($g = 10 \text{ N/kg}$)

Height = m

THIS QUESTION CONTINUES ON PAGE 16



- (iii) A data logger is then used to monitor the motion of the ball as it rises from the spring. A graph is automatically produced showing how the velocity of the ball changes with time. The graph is shown below.



- I. One student states that the height reached by the ball is different from that which you have calculated in (b)(ii). Show that this statement is true. [2]
Space for calculation.

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II. Explain why the height reached by the ball is different from the theoretical height. [2]

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III. Suggest **one** change that could be made that would enable a ball to reach a greater height in this experiment. [1]

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END OF PAPER



Surname	Centre Number	Candidate Number
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SCIENCE (Double Award)

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weight = mass $\times$ gravitational field strength	$W = mg$
work = force $\times$ distance	$W = Fd$
kinetic energy = $\frac{\text{mass} \times \text{velocity}^2}{2}$	$\text{KE} = \frac{1}{2}mv^2$
change in potential energy = mass $\times$ gravitational field strength $\times$ change in height	$\text{PE} = mgh$
force = spring constant $\times$ extension	$F = kx$
work done in stretching = area under a force-extension graph	$W = \frac{1}{2}Fx$

SI multipliers

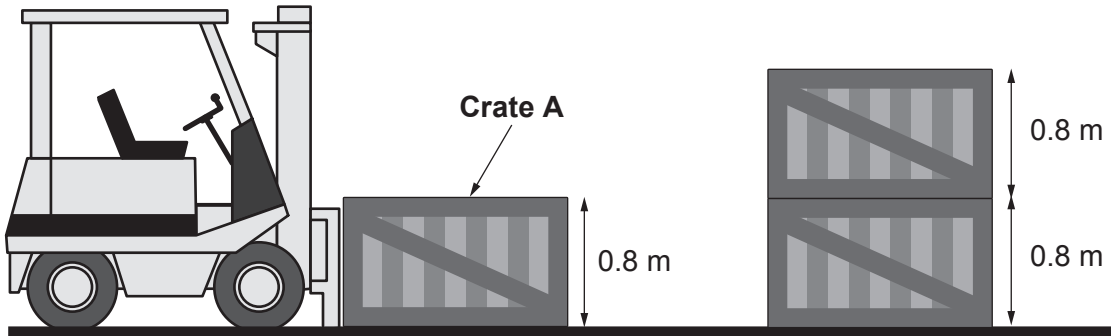
Prefix	Multiplier
p	1×10^{-12}
n	1×10^{-9}
μ	1×10^{-6}
m	1×10^{-3}

Prefix	Multiplier
k	1×10^3
M	1×10^6
G	1×10^9
T	1×10^{12}



Answer **all** questions.

1. On a farm a fork-lift truck is used to stack wooden crates of Welsh apples.



- (a) (i) Each crate of apples has a weight of 450 N and is 0.8 m high.
Use an equation from page 2 to calculate the work done in putting **crate A** on top of the **2 other** crates. [3]

Work done = J

- (ii) State the gain in potential energy (PE) of the crate when lifted on to the stack. [1]

Gain in PE = J

- (iii) Give a reason why the fork-lift truck uses more energy lifting crate A than the work done calculated in part (i). [1]

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.....

- (b) Each crate of apples has a weight of 450 N.
The mass of the empty crate is 12 kg.
Calculate the mass of apples contained in the crate.
(On Earth, an object of weight 10 N has a mass of 1 kg.) [2]

Mass of apples = kg



2. Jupiter has 79 moons in orbit around it. The table shows data for 4 of its moons.

Name of moon	Mean diameter (km)	Mean temperature (°C)	Orbit radius (km)	Orbit time (days)
Io	3 660	−163	421 700	1.8
Europa	3 120	−171	671 000	3.6
Ganymede	5 260	−163	1 070 400	7.2
Callisto	4 820	−139	1 882 700	16.7

Use information from the table above to answer the following questions.

- (a) (i) State which moon has the highest mean temperature. [1]
- (ii) For the solar system, as the orbit radius increases around the Sun, the mean temperature of the planets generally decreases.
Explain why the orbit radius of the moons around Jupiter does not affect their temperatures in the same way. [2]
-
-
-
- (iii) Peter states that as Callisto has the longest orbit time it must be the largest moon. Determine whether Peter’s claim is correct. [1]
-
-
- (iv) Peter correctly notices that Ganymede has an orbit time that is exactly double Europa’s.
Peter suggests that as the orbit time **doubles**, the orbit radius also **doubles**.
Use **only** the data for Ganymede and Europa to determine whether Peter’s claim is true. [1]
Space for calculations.
-
-



Examiner
only

- (b) (i) Two of the statements listed below are correct. One correct statement has already been ticked.
Tick (✓) one more box to show the other correct statement. [1]

An Astronomical Unit (AU) is the mean distance that separates the Earth and the Sun.

☒

A light year is a measurement of time.

☐

A light minute is the distance travelled by light in 60 seconds.

☐

A light year is smaller than a light second.

☐

- (ii) Both Earth and Jupiter travel in elliptical paths around the Sun.
As they orbit the Sun the closest distance between Jupiter and Earth is 588 000 000 km. This is equivalent to 3.92 AU.

Calculate the distance, in km, that separates the **Earth and the Sun**. [2]
(1 AU = Earth to Sun distance)

Distance = km

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Examiner
only

3. Carbon-12 ($^{12}_6\text{C}$) and carbon-14 ($^{14}_6\text{C}$) are isotopes.

(a) Explain, in terms of the **numbers of particles**, why carbon-14 has an unstable nucleus and carbon-12 has a stable nucleus. [3]

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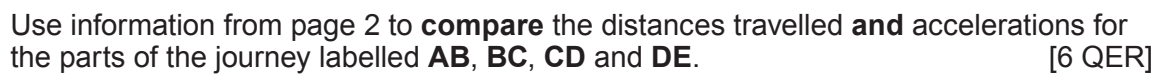
(b) An unstable nucleus becomes more stable following the emission of radiation. **Complete the table below.** [4]

Type of radiation	Symbol	Description
gamma		Electromagnetic (em) wave
alpha	$^4_2\alpha$	
beta		

7



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07



(b) The diagram shows a 5-carriage bi-modal electric/diesel train.



The table below shows information about two types of bi-modal electric/diesel trains.

Train	Mass ($\times 10^5$ kg)	Maximum speed (m/s)	Standard acceleration (m/s^2)	Standard deceleration (m/s^2)	Emergency deceleration (m/s^2)
5 carriage	2.3	55.8	0.7	1.0	1.2
9 carriage	4.4	55.8	0.7	1.0	1.2

For the journey from Swansea to London **two** of the 5-carriage trains are joined, making a 10-carriage train.
The 10-carriage train has the same speed and acceleration as a 5-carriage train.

(i) State Newton’s second law of motion as an equation. [1]

(ii) Use information from the table to answer the following questions.

I. Use an equation from page 2 to calculate the resultant force needed to accelerate the 10-carriage train. [2]

Resultant force = N

II. Use the equation:

acceleration = $\frac{\text{change in velocity}}{\text{time}}$

to calculate the time taken to accelerate the 10-carriage train from rest to its maximum speed. [2]

Time = s



Examiner
only

(iii) A student states that the 9-carriage train is approximately **double** the mass of the 5-carriage train and if they are both travelling at the same speed it will take approximately **double** the time to stop in an emergency.
Explain whether the **2 claims** made by the student are correct. [3]

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14

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09



5. An experiment was carried out by students to investigate the mean speed of a stack of 6 cake cases falling from different heights.
The weight of 6 identical cake cases was measured 5 times.
Their results are shown below in **Table 1** and there were no anomalies.

Table 1

Measurement	1	2	3	4	5
Weight of 6 cake cases (N)	0.036	0.032	0.033	0.034	0.030

- (a) (i) Calculate the mean weight of the 6 cake cases. [1]

Mean weight = N

- (ii) Use the equation:

$$\text{uncertainty} = \frac{\text{maximum value} - \text{minimum value}}{2}$$

to calculate the uncertainty in the mean weight of the 6 cake cases. [1]

Uncertainty in the mean = N

- (iii) The teacher states if the percentage uncertainty in the mean is **less than 10 %** the data is repeatable.

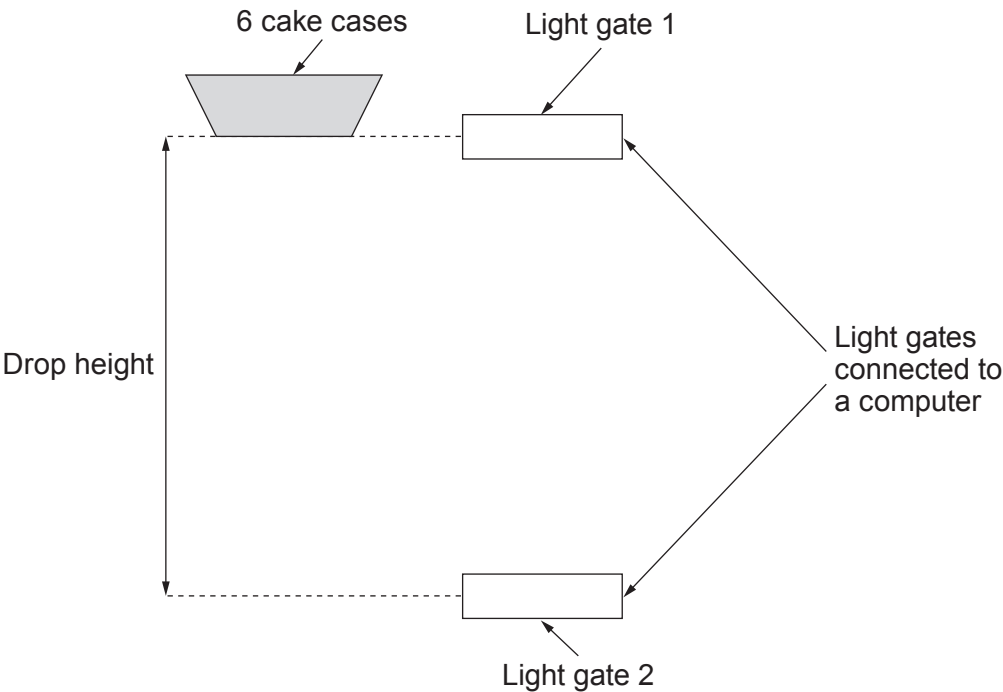
Use the equation:

$$\text{percentage uncertainty} = \frac{\text{uncertainty in the mean}}{\text{mean weight}} \times 100 \%$$

to determine if the data collected is repeatable. [2]



- (b) Light gates were used to time the falling cake cases. The vertical distance between the two light gates was adjusted so that different drop heights could be investigated. The apparatus is shown below.



Their results are shown in **Table 2** below.

Table 2

Drop height (m)	Time taken for the stack of 6 cake cases to fall (s)			Mean speed (m/s)
	Trial 1	Trial 2	Mean	
0.00				0.00
0.20	0.365	0.366	0.366	0.55
0.60	0.698	0.697	0.698	0.86
0.90	0.916	0.920	0.918	0.98
1.20	1.133	1.131	1.132	1.06
1.60	1.455	1.454	1.455	1.10
2.00	1.816	1.820	1.818	1.10
2.50	2.270	2.274	2.272	1.10
3.00	2.729	2.725	2.727	1.10



- (i) Give a reason why light gates, rather than a stopwatch, are used to measure the time taken for the stack of cake cases to fall. [1]

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- (ii) A student correctly notices that the mean speed of the cake cases at a drop height of 1.60 m is **double** the mean speed at a drop height of 0.20 m. The student states that the cake cases have **4 times** more kinetic energy after falling 1.60 m compared to 0.20 m.

Use an equation from page 2 to investigate whether the claim is correct.
The total mass of the 6 cake cases is 3.3×10^{-3} kg. [3]
Space for calculations.

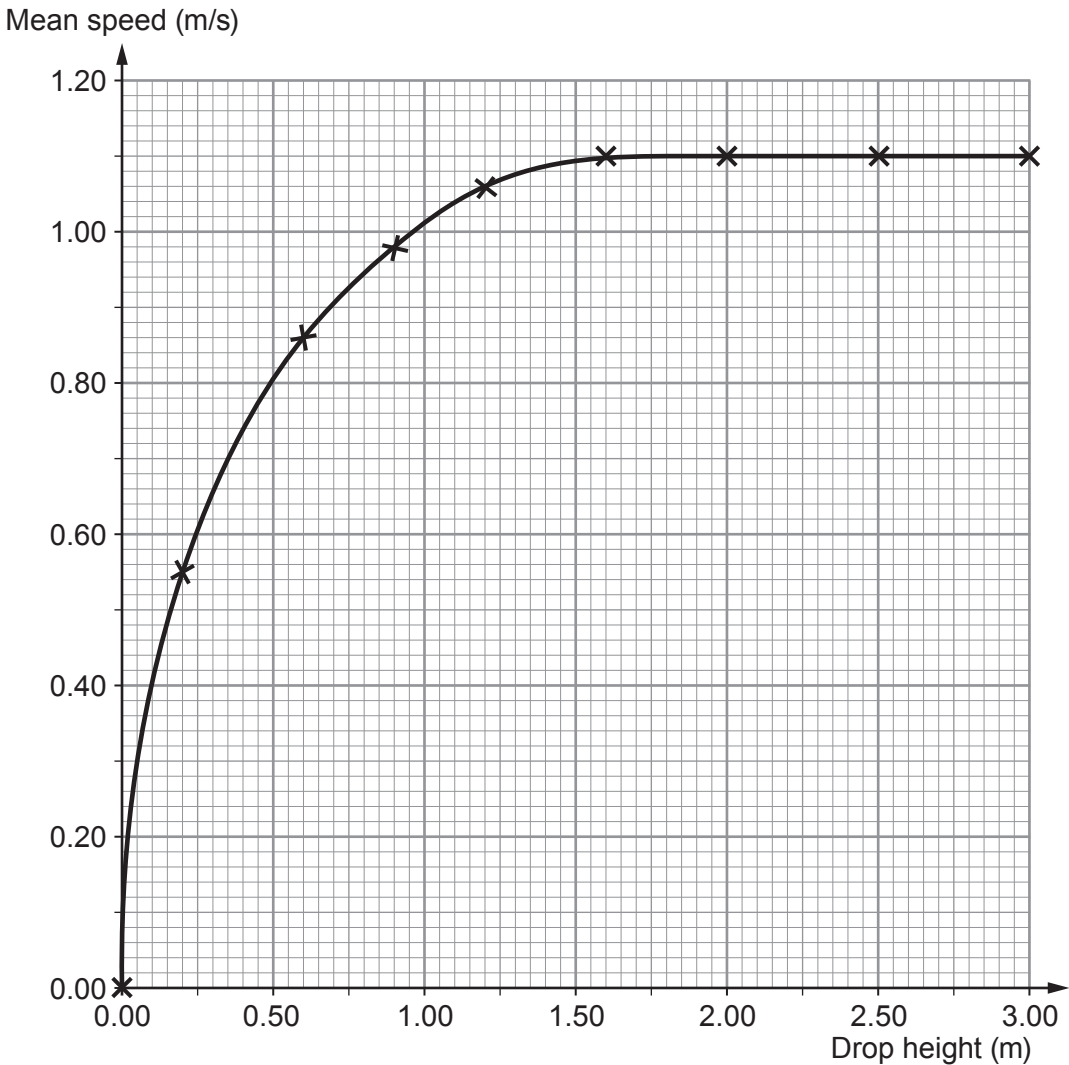
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Examiner
only

(c) A graph of mean speed against drop height for the 6 cake cases is shown below.



(i) Describe how the mean speed varies with drop height between **0.00 m and 1.50 m**. [2]

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.....

(ii) Use the graph to state the value of terminal speed for the cake cases. [1]

Terminal speed = m/s

(iii) I. Name the **two** forces acting on the falling cake cases. [1]

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II. Compare the size of these two forces when the cake cases fall at terminal speed. [1]

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13



6. Following the Chernobyl nuclear reactor accident in 1986, RIMNET (Radiation Incident Monitoring Network) was installed across the UK to monitor the consequences for the UK of nuclear incidents abroad.
GM tubes are used to measure background levels of radiation.
The permanent monitoring stations in Wales are located in the places shown on the map below.

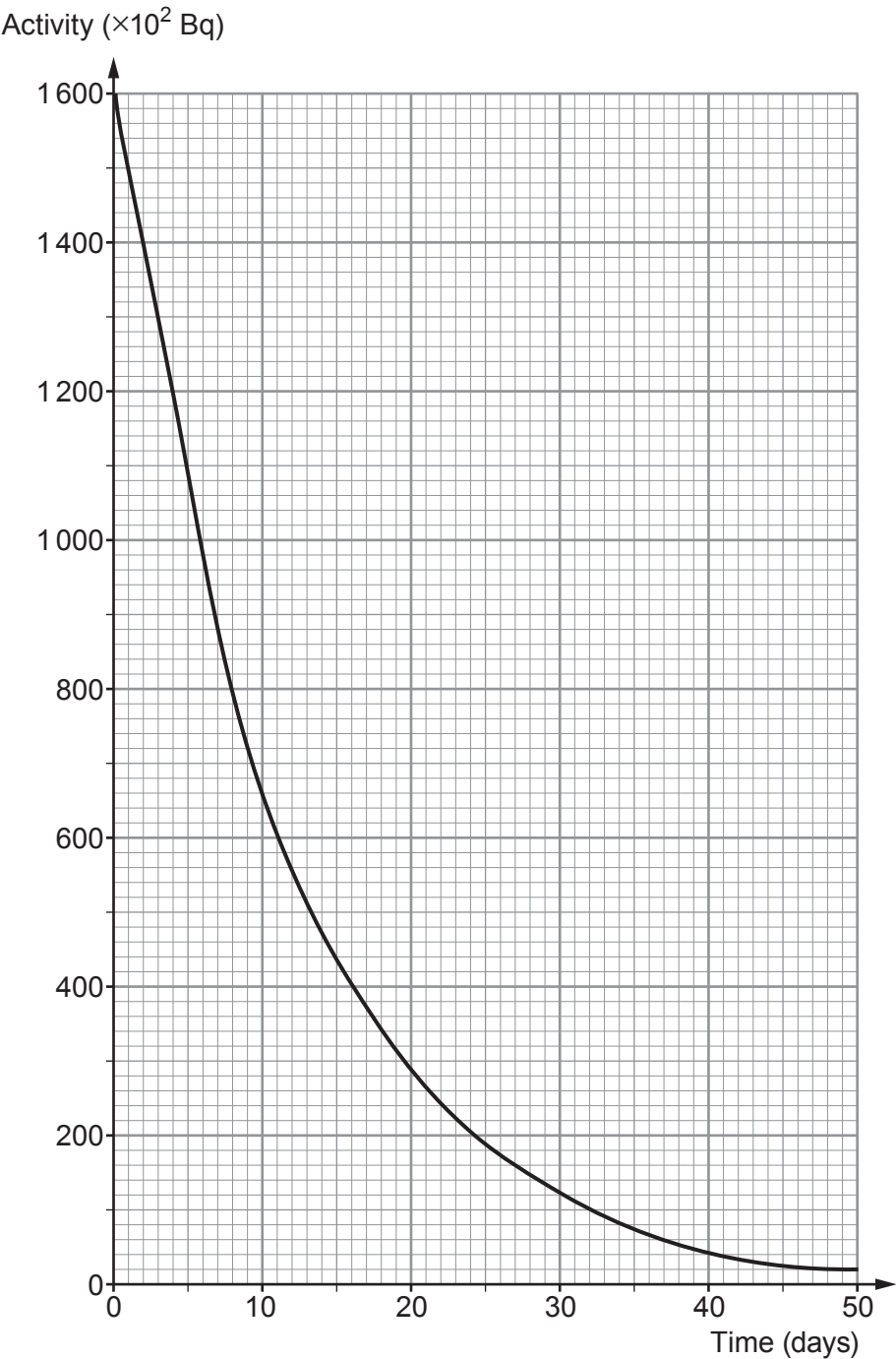


- (a) Suggest why there is **more than one** monitoring station located in Wales. [1]
-
- (b) The background count at one of the monitoring stations is 24 counts per minute.
Explain how, using a stopwatch and GM tube connected to a counter, an accurate value for the background count is determined. [2]
-
-
-
-



Examiner
only

- (c) Two radioisotopes released from the Chernobyl accident were iodine-131 (I-131) and caesium-137 (Cs-137).
The decay curve for a sample of I-131 is shown on the graph below.



- (i) State the meaning of half-life.

[2]

.....

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Examiner
only

(ii) Use information from the graph to determine the half-life of I-131. [1]

Half life = days

(d) A sample of Cs-137, with an activity of 1.5kBq, was obtained from near Chernobyl.
Calculate the time taken for this sample to reach a safe limit of $\frac{1}{512}$ of its original activity.
The half-life of Cs-137 is 30.2 years. [3]
Space for calculations.

Time taken = years

(e) Explain why in 2021, 35 years after the nuclear accident, scientists were concerned about the Cs-137 contamination in the Chernobyl area but not I-131. [2]

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END OF PAPER



Surname	Centre Number	Candidate Number
First name(s)		0



GCSE

3430UF0-1



THURSDAY, 25 MAY 2023 – MORNING

SCIENCE (Double Award)
Unit 6 – PHYSICS 2

HIGHER TIER

1 hour 15 minutes

For Examiner's use only		
Question	Maximum Mark	Mark Awarded
1.	5	
2.	10	
3.	12	
4.	10	
5.	10	
6.	13	
Total	60	

ADDITIONAL MATERIALS

In addition to this paper you will require a calculator and a ruler.

INSTRUCTIONS TO CANDIDATES

Use black ink or black ball-point pen. Do not use gel pen or correction fluid.
You may use a pencil for graphs and diagrams only.
Write your name, centre number and candidate number in the spaces at the top of this page.
Answer **all** questions.
Write your answers in the spaces provided in this booklet. If you run out of space, use the additional page(s) at the back of the booklet, taking care to number the question(s) correctly.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets at the end of each question or part-question.
The assessment of the quality of extended response (QER) will take place in question 4(a).



JUN233430UF0101

Equations

speed = $\frac{\text{distance}}{\text{time}}$	
acceleration [or deceleration] = $\frac{\text{change in velocity}}{\text{time}}$	$a = \frac{\Delta v}{t}$
acceleration = gradient of a velocity-time graph	
distance travelled = area under a velocity-time graph	
resultant force = mass $\times$ acceleration	$F = ma$
weight = mass $\times$ gravitational field strength	$W = mg$
work = force $\times$ distance	$W = Fd$
kinetic energy = $\frac{\text{mass} \times \text{velocity}^2}{2}$	$\text{KE} = \frac{1}{2} mv^2$
change in potential energy = mass $\times$ gravitational field strength $\times$ change in height	$\text{PE} = mgh$
force = spring constant $\times$ extension	$F = kx$
work done in stretching = area under a force-extension graph	$W = \frac{1}{2} Fx$

SI multipliers

Prefix	Symbol	Conversion factor	Multiplier
pico	p	divide by 1 000 000 000 000	1×10^{-12}
nano	n	divide by 1 000 000 000	1×10^{-9}
micro	μ	divide by 1 000 000	1×10^{-6}
milli	m	divide by 1000	1×10^{-3}
centi	c	divide by 100	1×10^{-2}

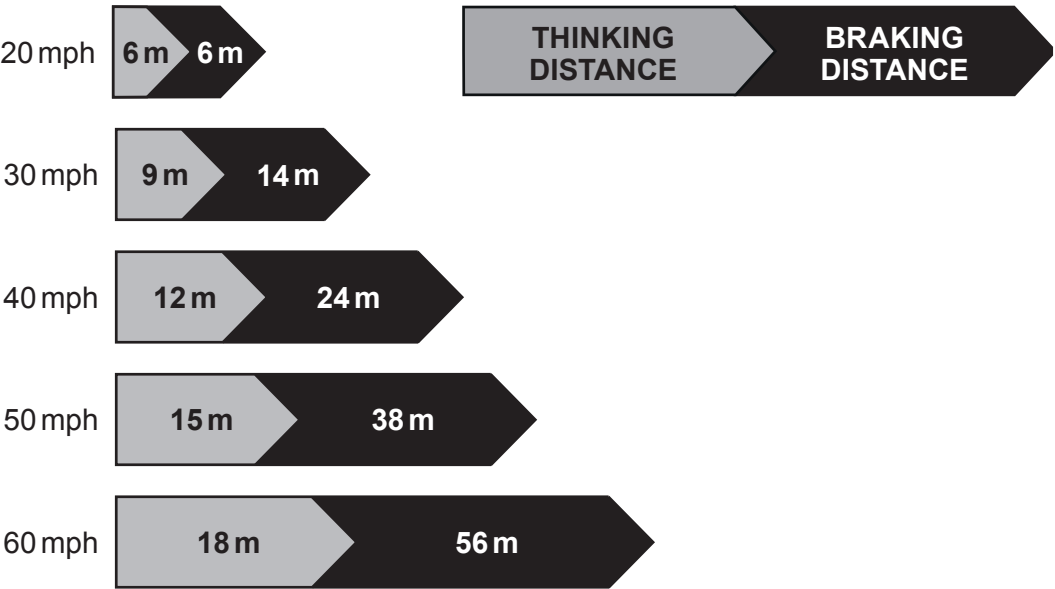
kilo	k	multiply by 1000	1×10^3
mega	M	multiply by 1 000 000	1×10^6
giga	G	multiply by 1 000 000 000	1×10^9
terra	T	multiply by 1 000 000 000 000	1×10^{12}



Answer **all** questions.

1. (a) When a car stops the overall stopping distance is made up of two distances: the thinking distance and the braking distance.
Increasing speed increases both the thinking distance and the braking distance.
- (i) State **one** factor, other than speed, which increases the thinking distance. [1]
-
- (ii) State **one** factor, other than speed, which increases the braking distance. [1]
-

(b) The diagram gives information about stopping distances at different speeds.



On a dangerous road, it is proposed to reduce the speed limit from 40 mph to 20 mph.

Bethan makes the following 3 suggestions.

1. The thinking distance will halve.
2. The braking distance will halve.
3. The overall stopping distance will halve.

Explain whether you agree with each suggestion.
Include data from the diagram to support your answer.

[3]

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2. (a) A class investigates radioactive decay. They model decay using **8-sided** dice.



Each of the 10 groups has 50 dice.
They throw the dice and remove all which land with an 8 facing upwards.
These represent decayed nuclei.
They repeat 7 more times and record the number of dice remaining after each throw.
Each group's results are then added together.

- (i) Freya suggests that it is good practice to add the results together to give a larger sample size.
Explain whether you agree. [1]

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.....

- (ii) The teacher calculates that after the first throw **around** 440 of the 8-sided dice should remain out of the 500.
Explain how she determined this number. [2]

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- (b) The results from the experiment are given in the table below.

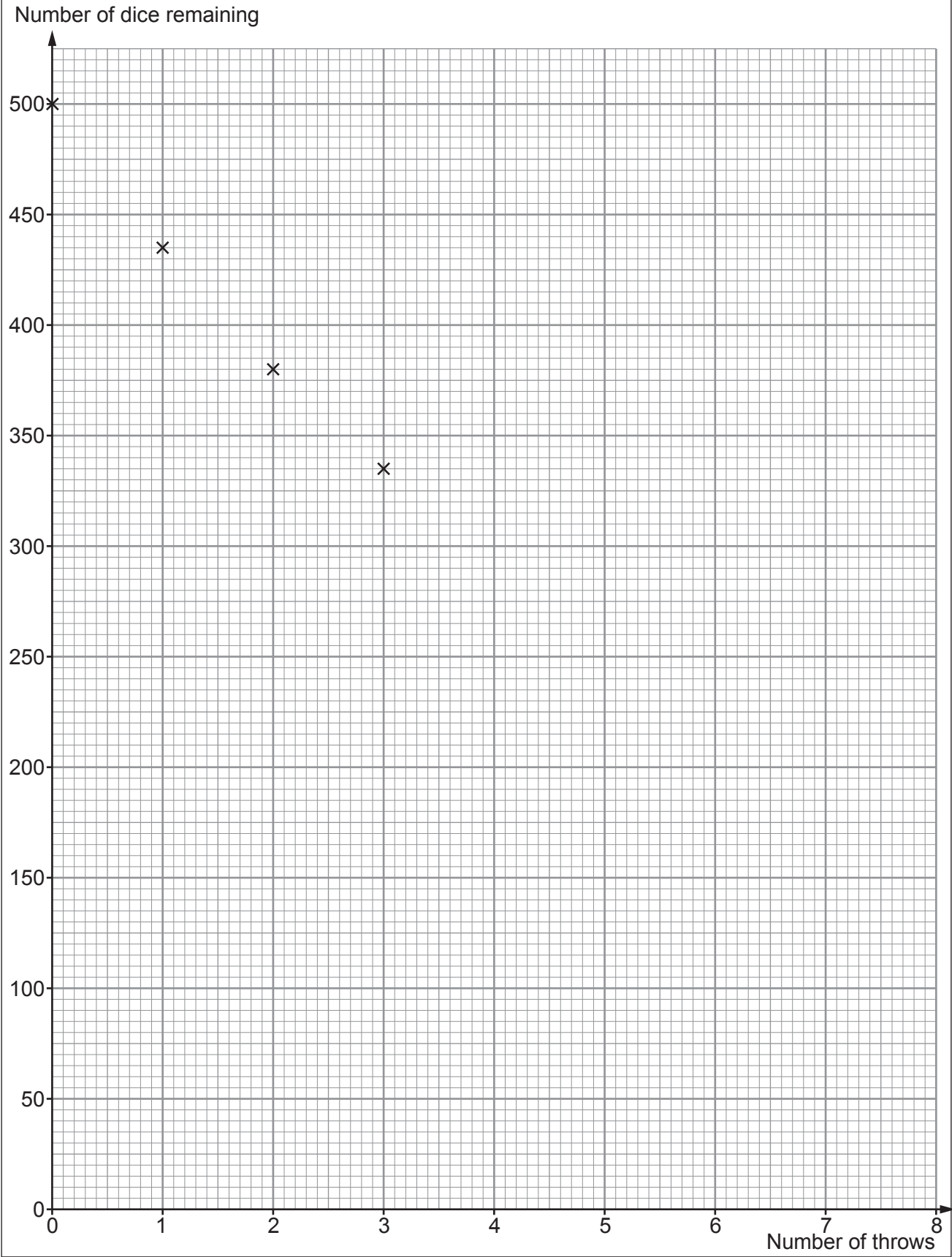
Number of throws	Number of dice remaining
0	500
1	435
2	380
3	335
4	291
5	256
6	224
7	196
8	171



Examiner
only

(i) Plot the data on the grid below and draw a suitable curve.
The first 4 points have been plotted for you.

[3]



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only

- (ii) **Add lines to the graph** to find the number of throws required to halve the number of dice. This is the half-life.
Give your answer to 1 decimal place. [2]

number of throws =

- (iii) The experiment is repeated with 10-sided dice.
leuan suggests the data could be used to model nuclear decay with a shorter half-life than with 8-sided dice.
Explain whether you agree. [2]

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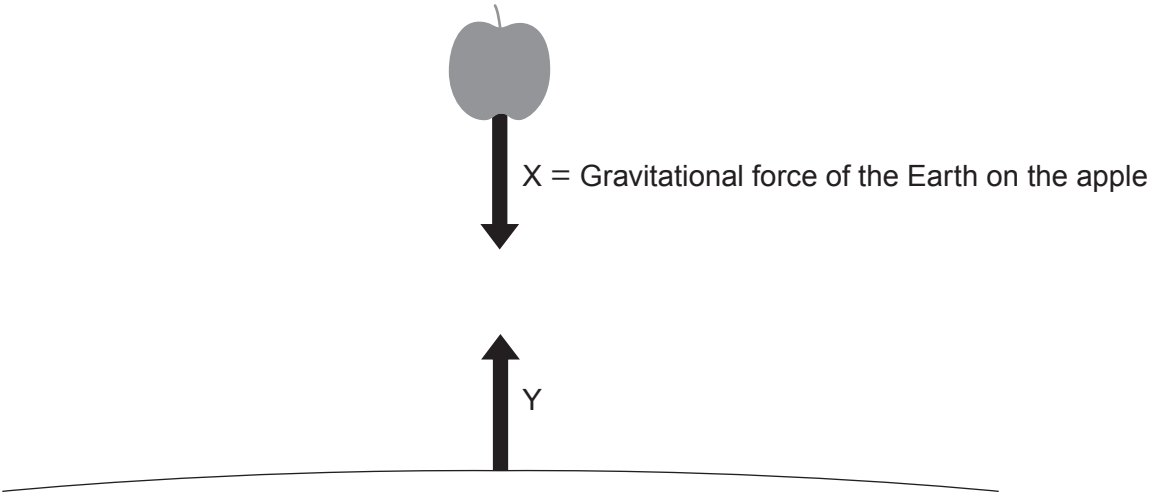
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3. Newton’s third law states that if a body A exerts a force on body B, body B exerts an equal and opposite force on body A.

(a) The diagram (not to scale) shows an apple of weight 1.5 N falling to the Earth.



(i) Underline the correct word in the brackets to describe force Y. [2]

Force Y is the (**gravitational** / magnetic / mass) force of the apple on the (air / **Earth** / Moon).

(ii) State how the two forces, X and Y, compare. [1]

.....

(iii) Use an equation from page 2 to calculate the mass of the apple. [2]
($g = 10\text{ N/kg} = 10\text{ m/s}^2$)

mass = kg

(b) The apple is dropped from rest.

(i) State the initial acceleration of the apple. [1]

acceleration = m/s^2



(ii) At some time later in the fall, the air resistance acting on the apple is 0.25 N.



Determine the size of the resultant force acting on the apple. [1]

resultant force = N

(iii) Use the equation:

resultant force = mass $\times$ acceleration

to calculate the new acceleration of the apple. [3]

acceleration = m/s^2

(c) The Apollo 15 astronaut David Scott performed an experiment on the Moon, where there is no air. He predicted that if a hammer and a feather were dropped together from the same height, they would hit the Moon's surface at the same time. Explain whether you agree with this prediction. [2]

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4. (a) Radioactive waste contains a mixture of isotopes with a variety of half-lives. The isotopes emit alpha, beta and gamma radiation. Discuss the safe storage of radioactive waste. Include in your answer:

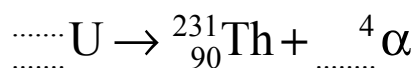
- a description of alpha, beta and gamma radiation,
- the properties of alpha, beta and gamma radiation,
- how radioactive waste could be safely stored.

[6 QER]

- (b) One isotope of uranium (U) decays naturally by alpha (α) decay into thorium (Th) which in turn decays into protactinium (Pa) by beta (β) decay.

Complete the decay equation for this decay.

[4]



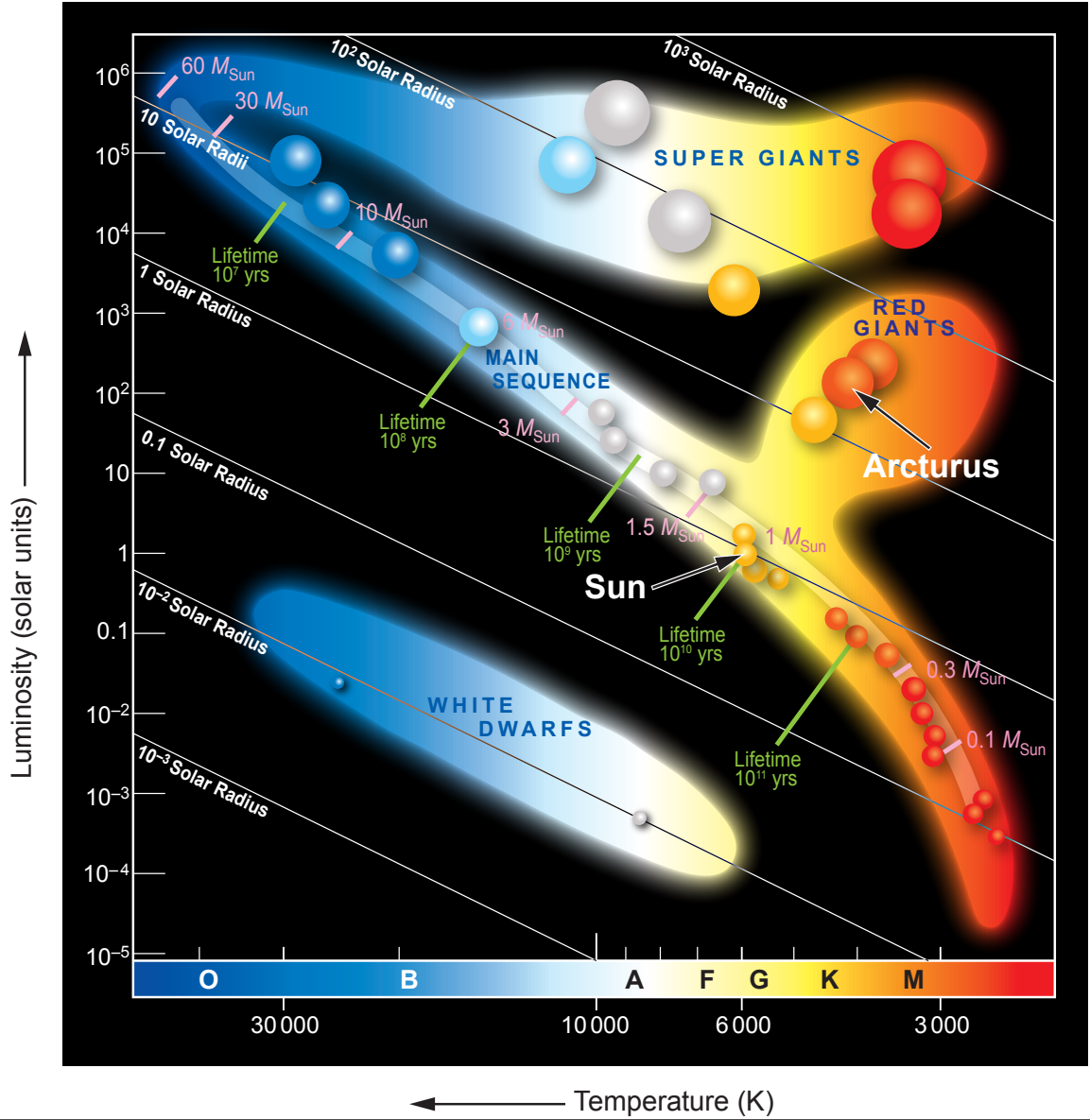
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Examiner
only

5. (a) Arcturus is a red giant star, 37 light-years away from Earth.
- (i) State how long light from Arcturus takes to reach the Earth. [1]
- time =
- (ii) Use your answer to (i) and an equation from page 2 to determine the distance of Arcturus from Earth in metres. [3]
- (1 year = 31 600 000 s, speed of light, $c = 3 \times 10^8$ m/s)
- distance = m

(b) The Hertzsprung-Russell (HR diagram) below shows the properties of different stars.



Examiner
only

- (i) The Sun and Arcturus have different colours.
Use information from the diagram to compare their other properties. [2]

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- (ii) The Sun is currently on the main sequence.
At the end of this stage of its life cycle it will become a red giant like Arcturus.
Explain, in terms of forces **and** fusion, why the Sun will become a red giant. [4]

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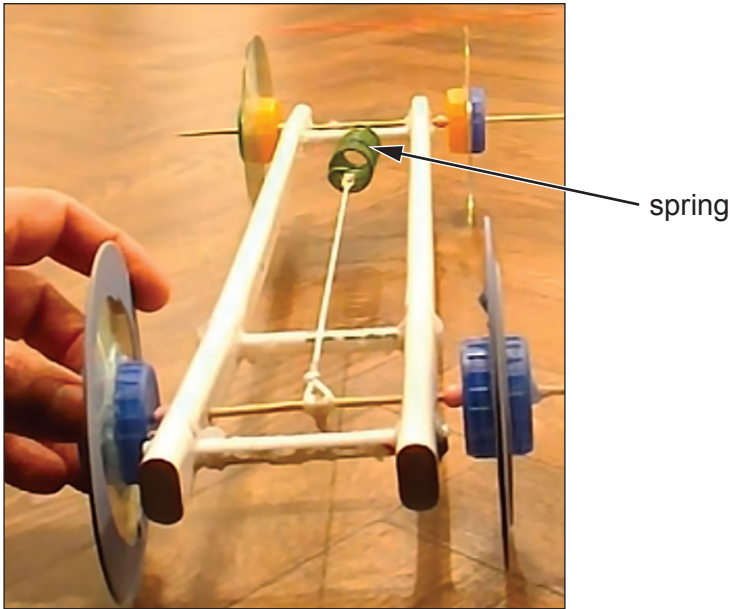
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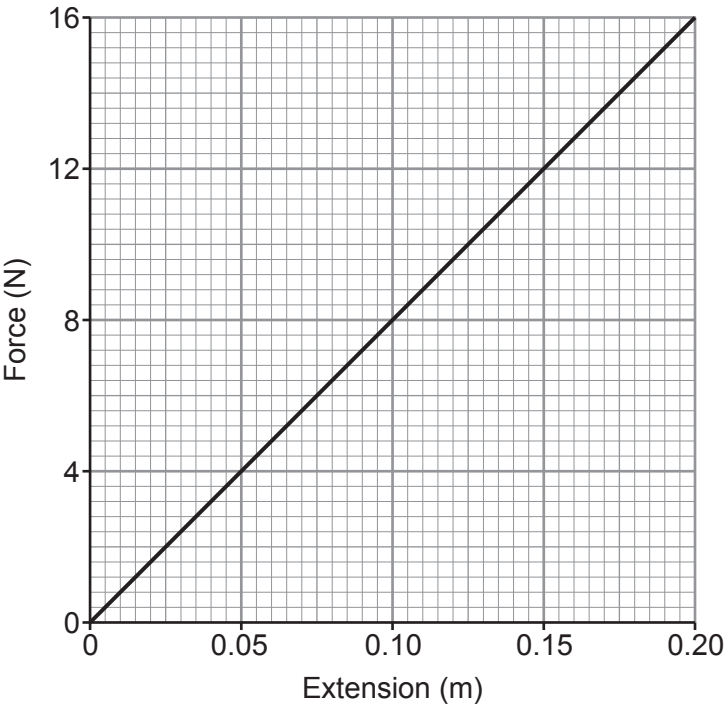
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6. Some students make a car powered by a spring.
The spring is stretched as the back wheels on the car are wound-up.



- (a) A force-extension graph for the spring used in the car is shown below.



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- (i) Use the graph and an equation from page 2 to calculate the work done in stretching the spring by 0.15 m. [2]

work done = J

- (ii) When the car is released the work done in stretching the spring is transferred to the car as kinetic energy.
Use your answer to part (i) and the equation:

kinetic energy = $\frac{1}{2} \times \text{mass} \times \text{velocity}^2$

to calculate the theoretical maximum velocity of the car if its mass is 0.075 kg.
Give the unit with your answer. [4]

velocity =

unit =

- (iii) Explain why, in practice, the car will not reach the theoretical maximum velocity. [2]

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TURN OVER FOR THE REST OF THE QUESTION



Examiner
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- (b) The car’s maximum kinetic energy is found to be 0.8 J.
A force brings it to rest in a distance of 1.5 m.

Use the equation:

$$\text{work done} = \text{force} \times \text{distance}$$

to determine the size of the force. [3]

force = N

- (c) The spring constant of the spring used in the toy car is 80 N/m.
Maddie suggests that it would be better to use a spring with a spring constant of 160 N/m because for the same extension, double the energy would be stored in the spring.
Explain whether you agree. [2]

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END OF PAPER

